Representation Learning & Generic Clustering Algorithms

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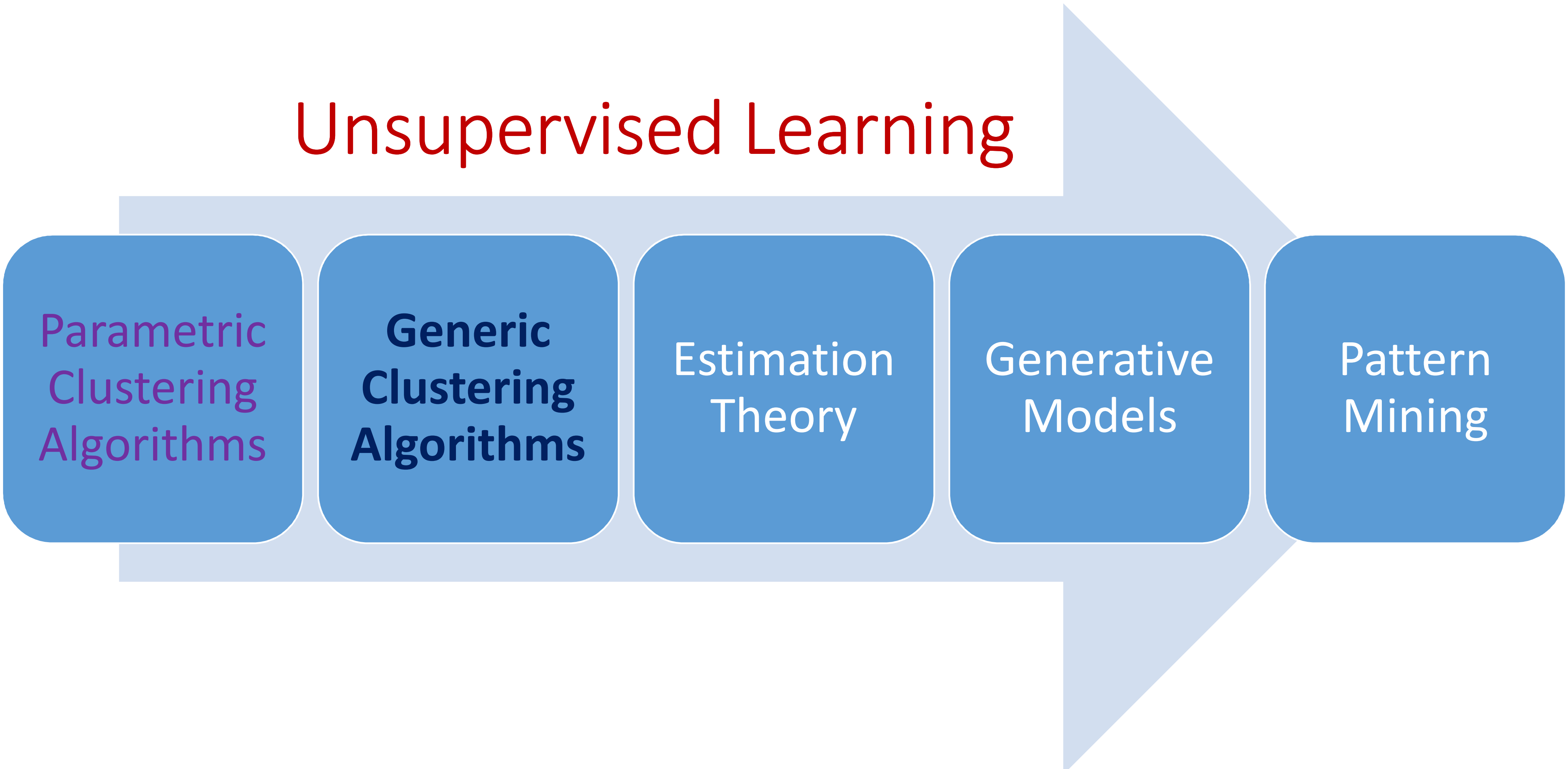
Unsupervised Learning
Parametric Clustering Algorithms
Generic Clustering Algorithms
Estimation Theory
Generative Models
Pattern Mining

Customer Purchase Patterns

#	Bread	Butter	Sugar	Tea	Coffee	Juice	Chocolate	Biscuit	Milk	Diaper	Salt	Rice	Wheat
1	1	1	1	1	0	2	0	1	1	0	1	0	1
2	2	2	1	2	0	1	0	2	1	0	1	0	2
3	2	2	1	1	0	2	0	1	1	0	1	0	1
4	4	2	1	1	0	1	0	3	1	0	1	0	0
5	0	2	1	1	0	4	0	1	1	0	1	1	1
6	2	2	0	1	0	6	0	1	1	0	1	0	1
7	1	2	0	1	0	8	0	1	1	0	1	0	0
8	2	2	0	1	1	1	2	1	1	0	1	0	0
9	3	1	0	1	1	0	1	1	1	1	1	0	0
10	1	1	2	1	0	0	1	1	1	1	1	1	1
11	4	1	2	1	0	0	2	3	1	1	1	1	0
12	0	0	2	2	1	1	1	1	1	1	1	1	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Two Transactions

Distance Between Sets

$$\boldsymbol{S_1} = \{\boldsymbol{a}_i ; i = 1, \ldots n\}$$

$$\boldsymbol{S_2} = \{\boldsymbol{b}_j ; j = 1, \ldots m\}$$

$$\boldsymbol{d} : \boldsymbol{S_1} \times \boldsymbol{S_2} \rightarrow \mathbb{R}^+ \cup \{0\}$$

$$\boldsymbol{d}(\boldsymbol{a}_i, \boldsymbol{b}_j)$$

$$\boldsymbol{D_{min}}(\boldsymbol{S_1}, \boldsymbol{S_2}) = \min_{\boldsymbol{i,j}} \boldsymbol{d}(\boldsymbol{a}_i, \boldsymbol{b}_j)$$

$$\boldsymbol{D_{avg}}(\boldsymbol{S_1}, \boldsymbol{S_2}) = \frac{1}{mn} \sum_{\boldsymbol{i,j}} \boldsymbol{d}(\boldsymbol{a}_i, \boldsymbol{b}_j)$$

$$\boldsymbol{D_{max}}(\boldsymbol{S_1}, \boldsymbol{S_2}) = \max_{\boldsymbol{i,j}} \boldsymbol{d}(\boldsymbol{a}_i, \boldsymbol{b}_j)$$

$\boldsymbol{d}($  ,  $)$

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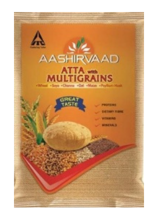
, 

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Distance Between Sets: Examples

$$S_1 = \{0, 1, 9\}$$

$$S_2 = \{3, 7\}$$

$$D_{min}(S_1, S_2) = \min_{i,j} d(a_i, b_j) = 2$$

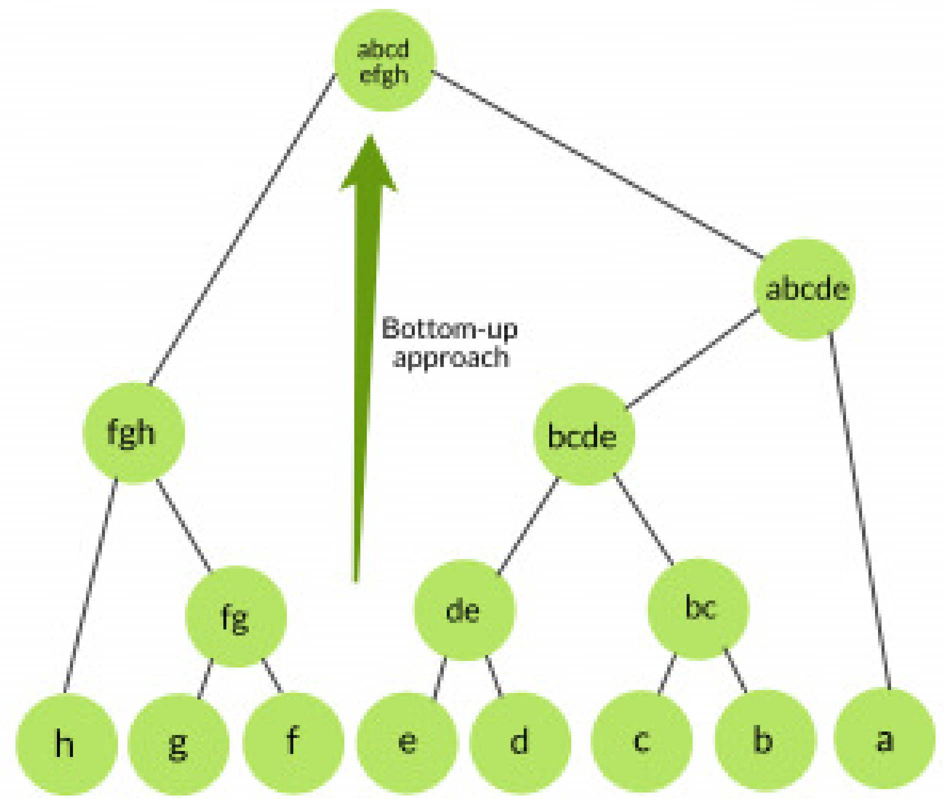
$$D_{avg}(S_1, S_2) = \frac{1}{mn} \sum_{i,j} d(a_i, b_j) = 4.33$$

$$D_{max}(S_1, S_2) = \max_{i,j} d(a_i, b_j) = 7$$

Distance Matrix

	0	1	9
3	3	2	6
7	7	6	2

Hierarchical Agglomerative Clustering

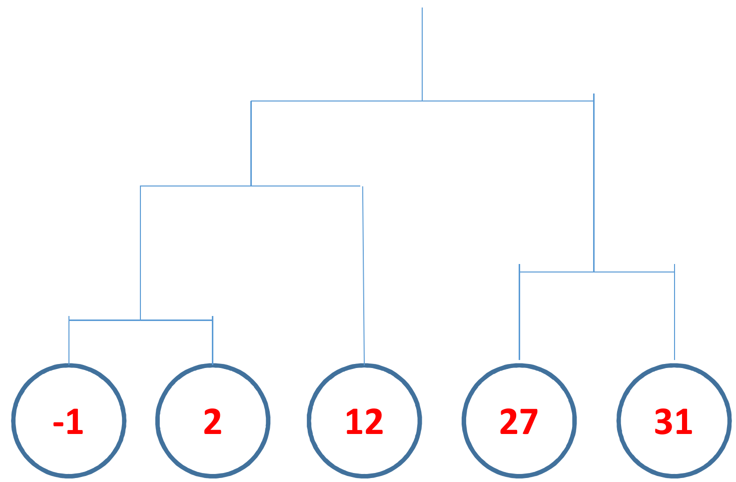


HAC: Dendrogram & Types

$\boldsymbol{S} = \{-1, 2, 12,27,31\}$

Distance Matrix

	-1	2	12	27	31
-1	0	3	13	28	32
2		0	10	25	29
12			0	15	19
27				0	4
31					0

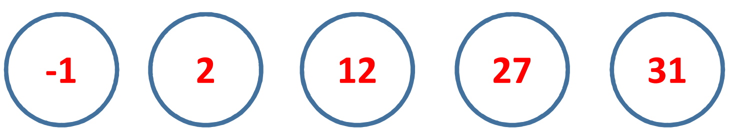


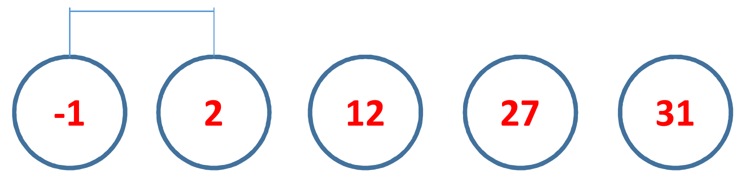
Dendrogram

- Single/Minimum Linkage $(\boldsymbol{D_{min}})$
- Average Linkage $(\boldsymbol{D_{avg}})$
- Maximum Linkage $(\boldsymbol{D_{max}})$

Illustrating Single Linkage HAC

	-1	2	12	27	31
-1	0	**3**	13	28	32
2		0	10	25	29
12			0	15	19
27				0	4
31					0

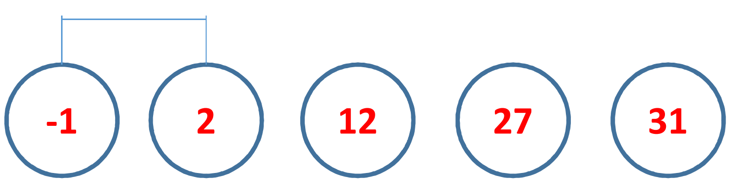


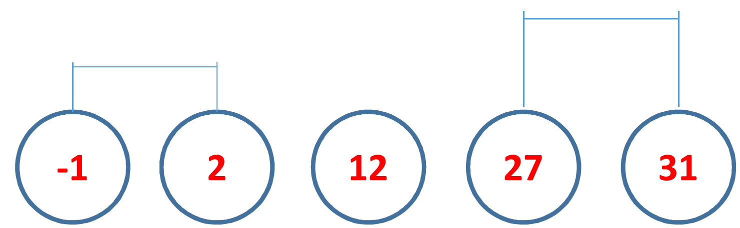


	{-1,2}	12	27	31
{-1,2}	0	10	25	29
12		0	15	19
27			0	4
31				0

Illustrating Single Linkage HAC

	{-1,2}	12	27	31
{-1,2}	0	10	25	29
12		0	15	19
27			0	**4**
31				0

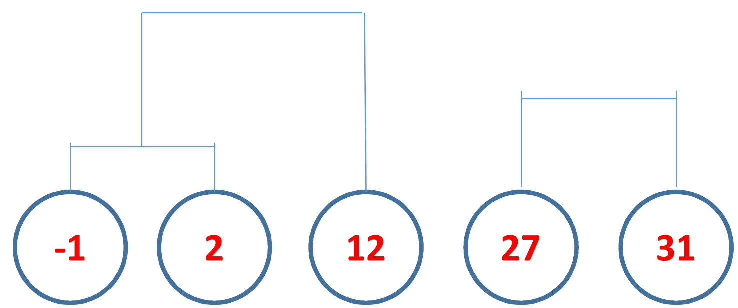


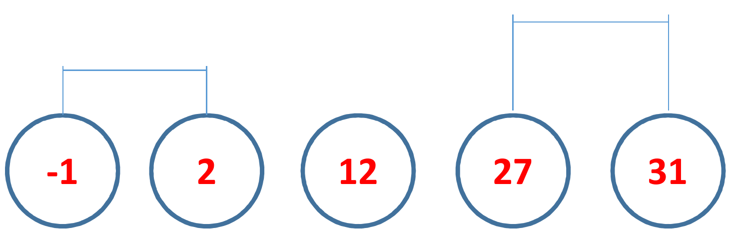


	{-1,2}	12	{27,31}
{-1,2}	0	10	25
12		0	15
{27,31}			0

Illustrating Single Linkage HAC

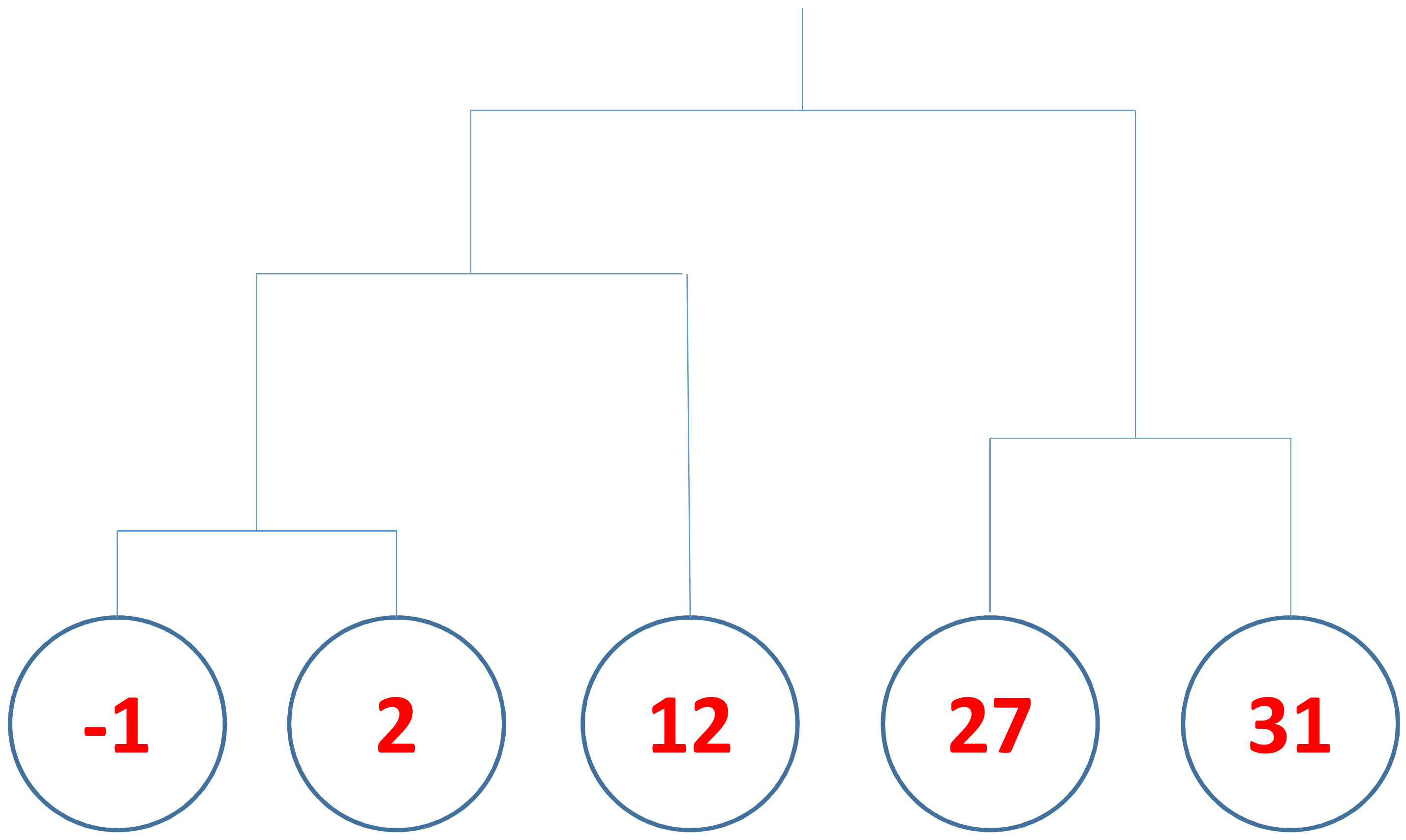
	{-1,2}	12	{27,31}
{-1,2}	0	**10**	25
12		0	15
{27,31}			0





	{-1,2,12}	{27,31}
{-1,2,12}	0	**15**
{27,31}		0

The Dendrogram

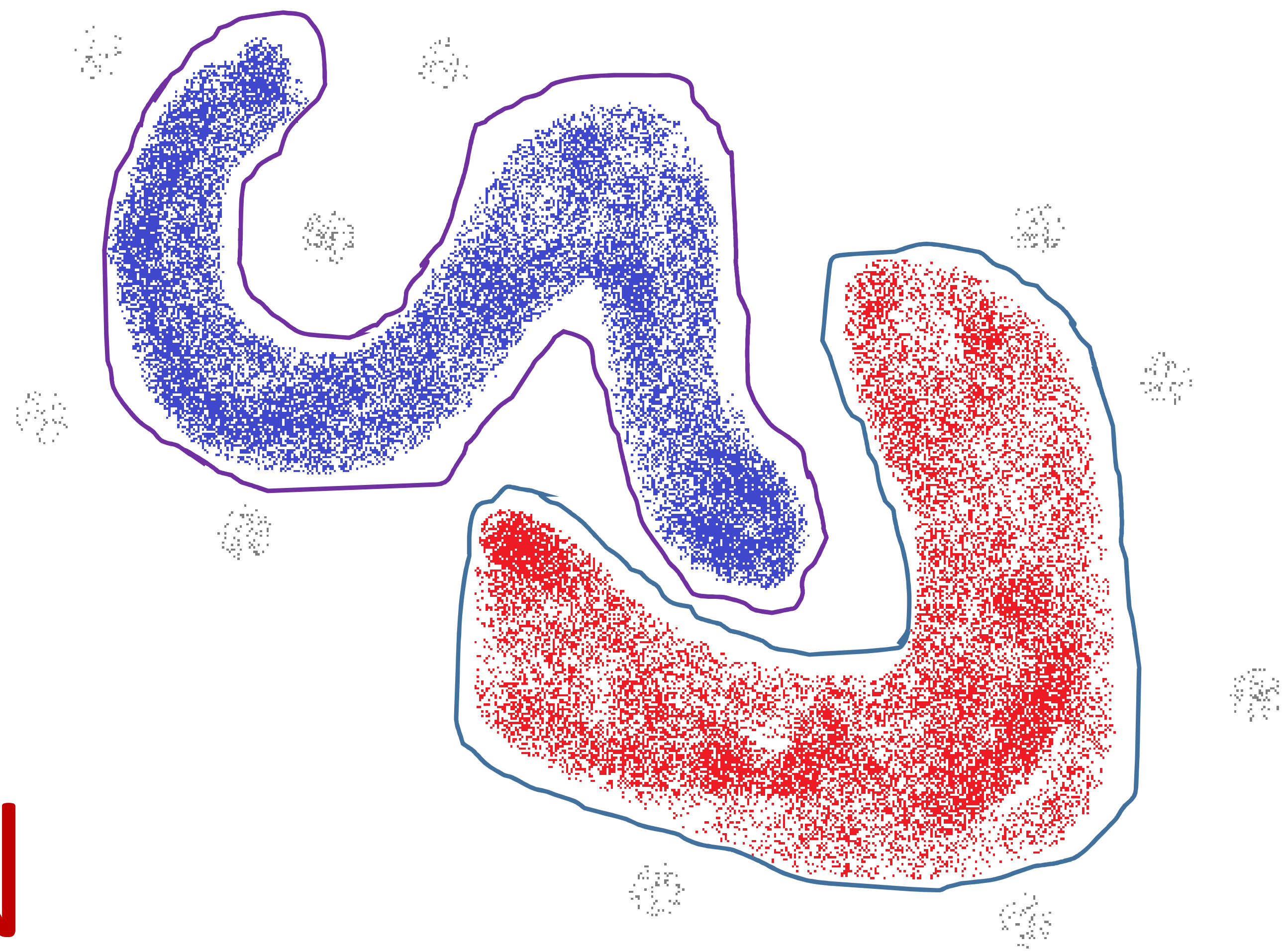


HAC: Advantages

- Apriori Information of Number of Clusters Not Required
- Works Well for Non-Convex Clusters
- Discovers Hierarchical Relation Among Clusters
- Discovered Hierarchy May Correspond to Meaningful Taxonomy
- For Non-Metric Data, Only Distance Matrix is Required
- Ease of Implementation

HAC: Drawbacks

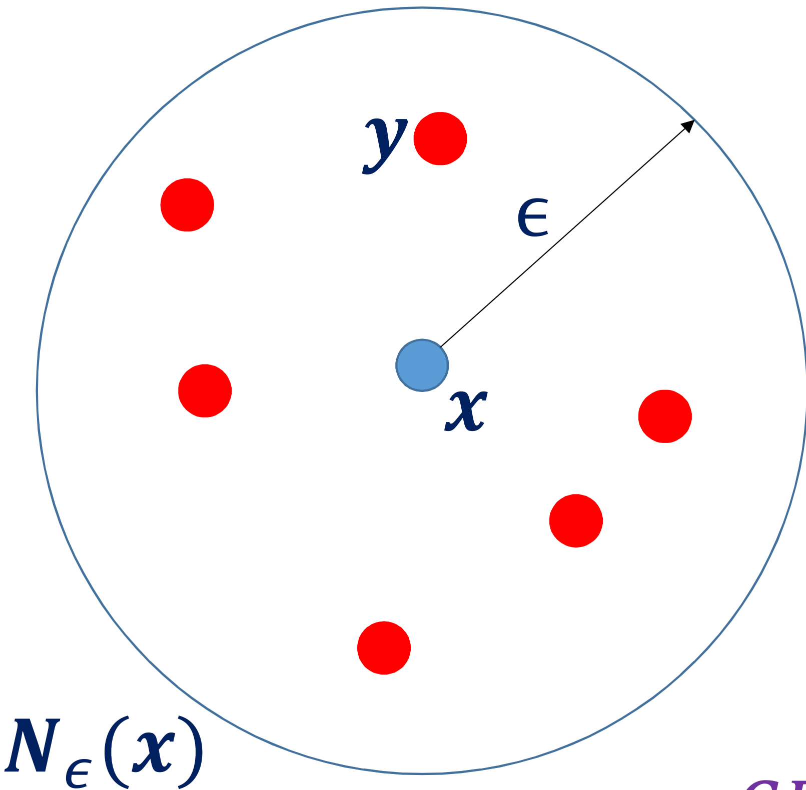
- Specification of Number of Clusters Not Easy from Dendrogam
- Very Time Consuming for Large Datasets



DBSCAN

Density Based Spatial Clustering for Applications with Noise

Spatial Density & Connectivity of Points



Direct Connectivity

$$DC(\boldsymbol{x},\boldsymbol{y};\epsilon) \Rightarrow [\|\boldsymbol{y}-\boldsymbol{x}\|_2 \le \epsilon]$$

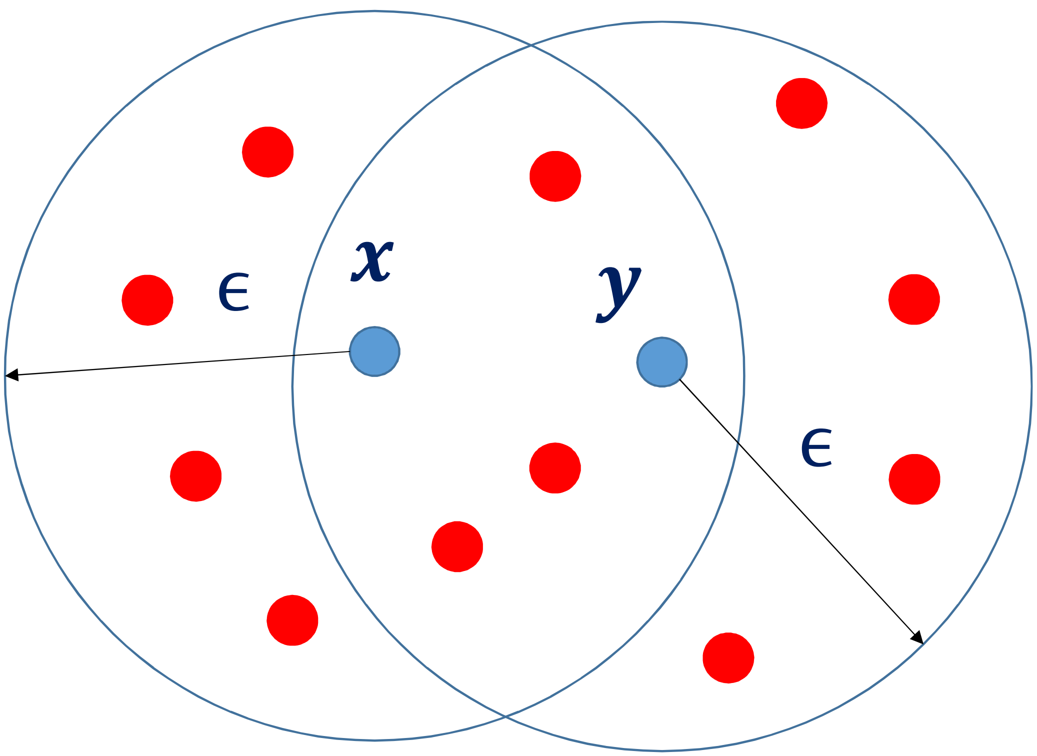
$$DC(\boldsymbol{x},\boldsymbol{y};\epsilon) \Rightarrow DC(\boldsymbol{y},\boldsymbol{x};\epsilon)$$

Core Point

$$N_\epsilon(\boldsymbol{x}) = \{\boldsymbol{z}: \|\boldsymbol{z}-\boldsymbol{x}\|_2 \le \epsilon\}$$

$$CP(\boldsymbol{x};\epsilon,\rho) \Rightarrow [|N_\epsilon(\boldsymbol{x})| \ge \rho]$$

Direct Density Connectivity & Reachability



$CP(\boldsymbol{x}; \epsilon, \rho)$ AND

$CP(\boldsymbol{y}; \epsilon, \rho)$

AND

$DC(\boldsymbol{x}, \boldsymbol{y}; \epsilon)$

$\Downarrow$

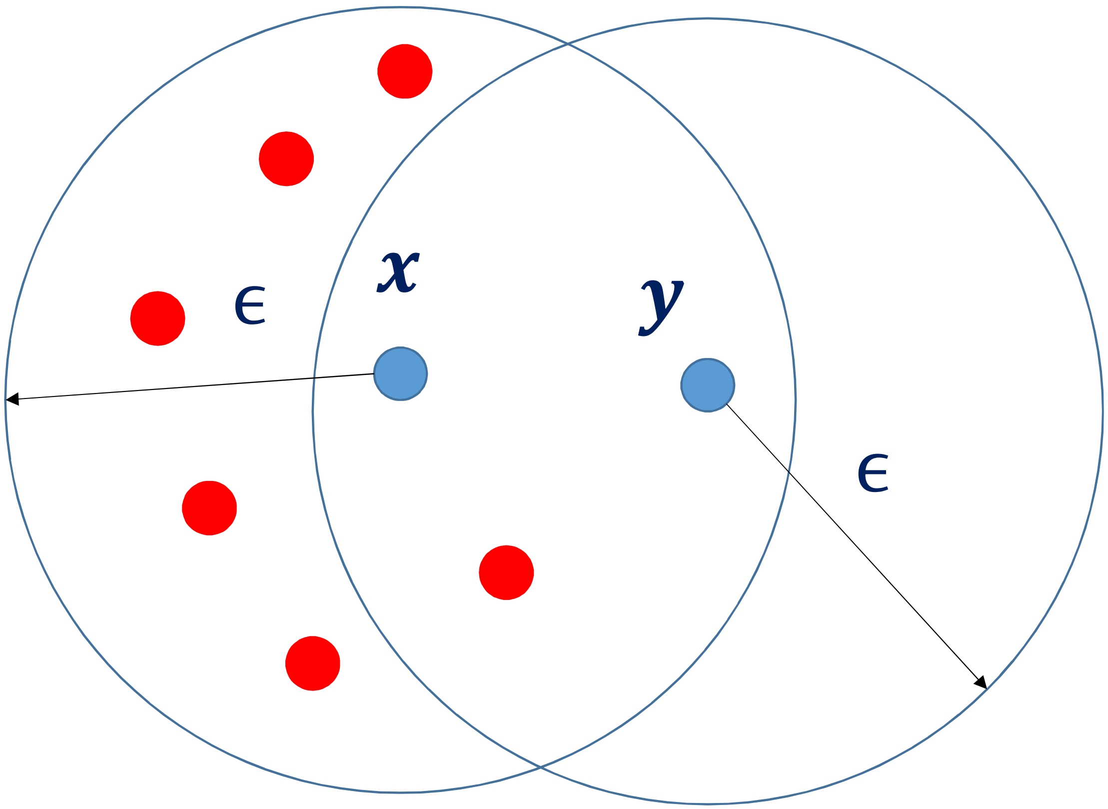
$\boldsymbol{DDR(x, y; \epsilon, \rho)}$ AND

$\boldsymbol{DDR(y, x; \epsilon, \rho)}$

DDR is Symmetric for Core Points Only

DDR is Asymmetric in General

Direct Density Connectivity & Reachability



$\boldsymbol{y}$ is a Border Point

$CP(\boldsymbol{x};\epsilon,\rho)$ AND

$\neg CP(\boldsymbol{y};\epsilon,\rho)$

AND

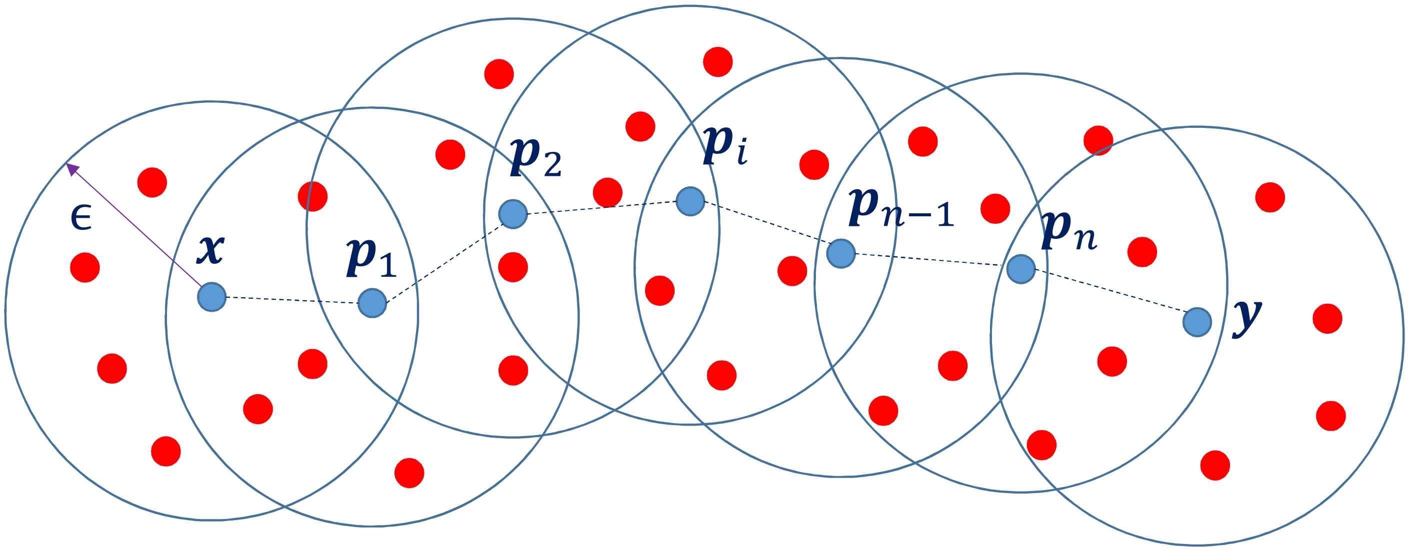
$DC(\boldsymbol{x},\boldsymbol{y};\epsilon)$

$\Downarrow$

$\boldsymbol{DDR}(\boldsymbol{x},\boldsymbol{y};\boldsymbol{\epsilon},\boldsymbol{\rho})$ AND

$\neg\boldsymbol{DDR}(\boldsymbol{y},\boldsymbol{x};\boldsymbol{\epsilon},\boldsymbol{\rho})$

Density Connectivity & Reachability

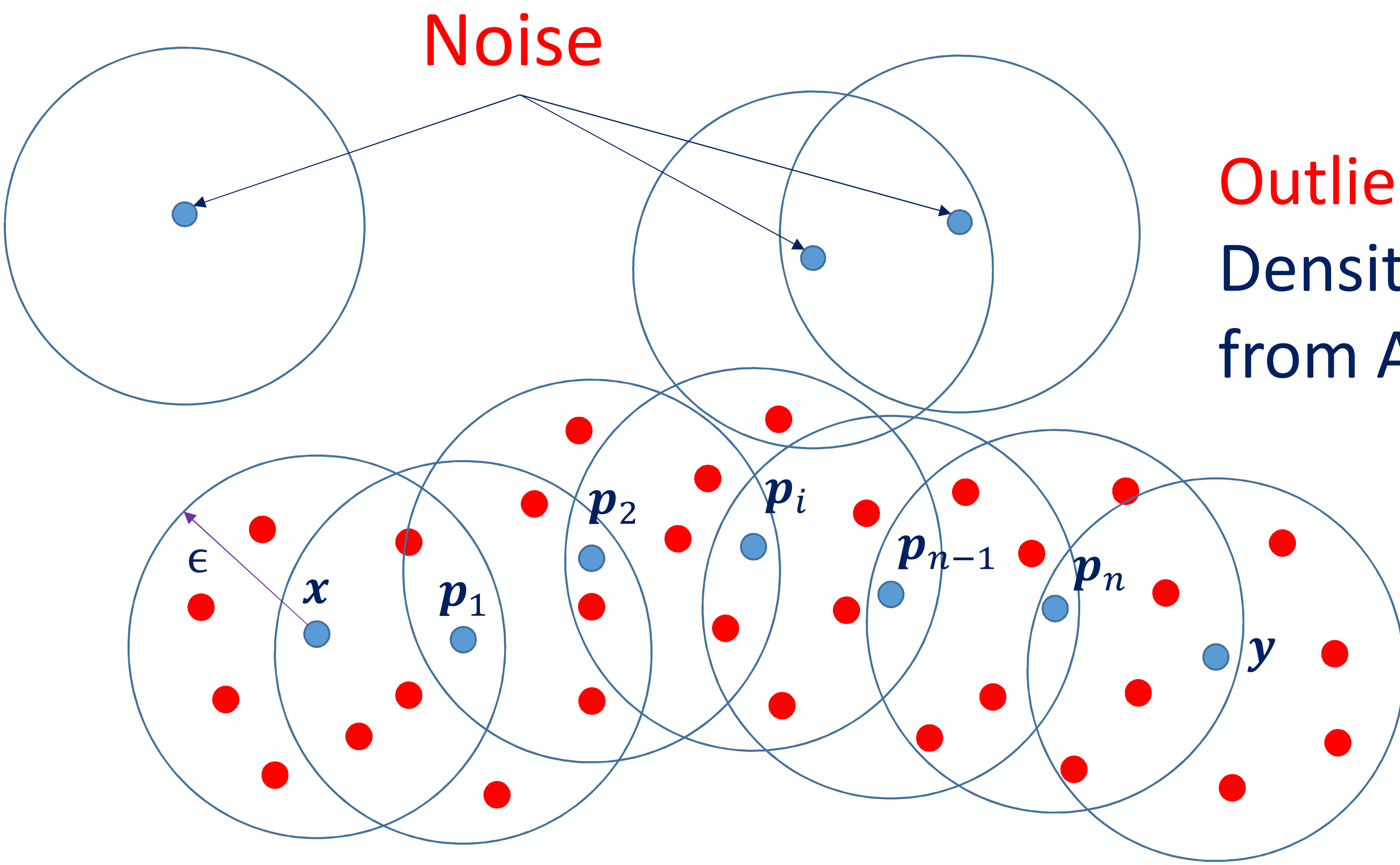


$DR(\boldsymbol{x}, \boldsymbol{y}; \epsilon, \rho)$

$\exists \boldsymbol{p}_i \; DDR(\boldsymbol{p}_i, \boldsymbol{p}_{i+1}; \epsilon, \rho); i = 1, \ldots (n-1)$ AND

$DDR(\boldsymbol{x}, \boldsymbol{p}_1; \epsilon, \rho)$ AND $DDR(\boldsymbol{p}_n, \boldsymbol{y}; \epsilon, \rho)$

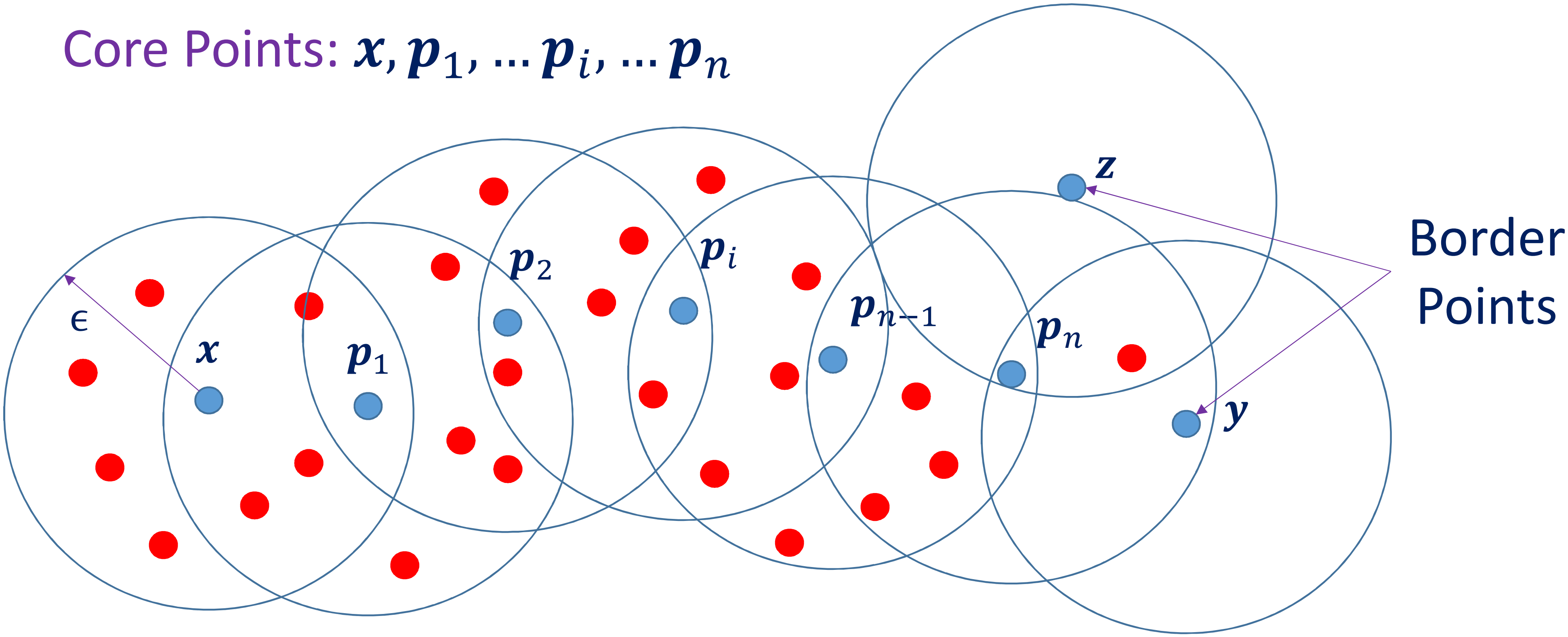
Noise or Outliers



Outliers: Not Density Reachable from Any Core Point

Cluster: Core & Border Points

Core Points: $\boldsymbol{x}, \boldsymbol{p}_1, \ldots \boldsymbol{p}_i, \ldots \boldsymbol{p}_n$



Density Reachable Core & Border Points Form a Cluster

- INPUT: Data $\boldsymbol{S} = \{\boldsymbol{a}_1, \ldots \boldsymbol{a}_i, \ldots \boldsymbol{a}_n\}$; Distance Matrix $\boldsymbol{D}$
- SET $\boldsymbol{k} = \mathbf{0}$; WHILE $\boldsymbol{S} \neq \boldsymbol{\varphi}$
    1. INCREMENT: $k = k + 1$; INITIALIZE Cluster: $\mathbf{C}_\mathrm{k} = \boldsymbol{\varphi}$; Core Points: $\boldsymbol{cpNum} = \mathbf{0}$;
    2. Randomly Choose First Point $\boldsymbol{a}$ from $\boldsymbol{S}$
    3. INITIALIZE FIFO Queue $\boldsymbol{Q} = \{\boldsymbol{a}\}$
    4. WHILE $\boldsymbol{Q} \neq \boldsymbol{\varphi}$
        A. EXTRACT: First Element of Queue: $\boldsymbol{\alpha} = \boldsymbol{Q}[\mathbf{1}]$
        B. EXTRACT: Neighborhood of $\boldsymbol{\alpha}$: $\boldsymbol{N}_\epsilon(\boldsymbol{\alpha}) = RegionQuery(\boldsymbol{\alpha}, \boldsymbol{D}; \epsilon)$
        C. PUSH UNFLAGGED Elements of $\boldsymbol{N}_\epsilon(\boldsymbol{\alpha})$ to $\boldsymbol{Q}$
        D. FLAG: Elements of $\boldsymbol{N}_\epsilon(\boldsymbol{\alpha})$ pushed to $\boldsymbol{Q}$
        E. MARK $\boldsymbol{\alpha}$ as either of CORE or BORDER Point and ASSIGN Label $\boldsymbol{k}$
        F. PUSH $\boldsymbol{\alpha}$ to $\boldsymbol{C}_k$; IF $\mathrm{CP}(\alpha)$ THEN $\boldsymbol{cpNum} = \boldsymbol{cpNum} + \mathbf{1}$
    5. END WHILE
    6. FLUSH $\boldsymbol{Q}$; UPDATE: $\boldsymbol{S} \leftarrow \boldsymbol{S} - \boldsymbol{C}_k$
    7. IF $\boldsymbol{cpNum} = \mathbf{0}$ THEN MARK $\boldsymbol{C}_k$ As NOISE
- END WHILE

DBSCAN: Algorithm

Advantages

- Works Very Well on Non-convex Datasets
- Can Detect Noise and Robust to Outliers
- Does Not Require the Number of Clusters
- Works on Both Numerical & Non-numerical Data
- Works Faster with Availability of Region Query

Disadvantages

- Algorithm Success Depends on Distance Measure
- Choice of ϵ Depends on Distance Measure
- Wide Variations in Spatial Density Lead to Algorithm Failure
- Depends on Distance Measure and Spatial Density
- Choice of (ρ, ϵ) can fail if Data is Not Well Understood

Summary

- Representation Learning using Spherical Clustering
- Representation Learning using Bag of Words
- Distances between Distributions
- Hierarchical Agglomerative Clustering
- DBSCAN

Thank You